MT-AIML, Honors AIML, Sem 5, Module 4 — Collocation Mining & Association Scoring, Problem 17

Architecture and Methodology

Application context. Executive voice-memo tools turn a dictated note or a council hearing into tags. The useful tags are not the most frequent words. They are fixed multiword expressions: role titles, procedural formulas, policy terms, and place names. This note specifies the collocation engine that proposes those phrases, the association scores it compares, and the decisions that change what the scores mean.

What counts as a collocation

A collocation here is a contiguous bigram or trigram that speakers treat as a unit. The unit can be institutionalized (city council, general fund), procedural (motion to, president pro tem), or a fixed name (long beach, mile high stadium). Compositional grammar such as of the is frequent and is not a collocation. A fixed discourse formula such as thank you is a collocation of a different kind: real, and not a domain tag. The gold file records the domain and procedural units the evaluation is allowed to count as hits. It is a curated sample, not a complete inventory of the corpus.

Association measures

All eight from-scratch scores share one 2×2 table. N is the number of sentence-internal bigram tokens, not the number of word tokens. On this corpus that N is 61728. NLTK's finder.N is 67046, which equals the processed token count. The difference is 5318, the same as the sentence count 5318: each sentence contributes one word that does not start a bigram. f(w1) is the number of bigrams that start with w1, and f(w2) is the number that end with w2. Raw unigram counts would count a sentence-final token in a slot it never occupies, and the bottom-right cell could go negative.

$O_{11} = f(w_1, w_2)$, $O_{12} = f(w_1) - O_{11}$, $O_{21} = f(w_2) - O_{11}$, $O_{22} = N - O_{11} - O_{12} - O_{21}$.

Maximum-likelihood probabilities use that same N: $P(w_1, w_2) = O_{11} / N$, $P(w_1) = (O_{11} + O_{12}) / N$, $P(w_2) = (O_{11} + O_{21}) / N$.

Pointwise mutual information

$PMI(w_1, w_2) = \log_2(P(w_1, w_2) / (P(w_1) P(w_2))) = \log_2(O_{11} \cdot N / (R_1 \cdot C_1))$.

NPMI = PMI / $(-\log_2 P(w_1, w_2))$. The denominator is zero when the joint probability is 0 or 1, and the score is then left undefined.

$PMI^k = (\log_2(P(w_1, w_2) / (P(w_1) P(w_2))))^k$, implemented as PMI ** k for k = 2 and k = 3. An even power turns a strong negative PMI into a large positive score.

PMI is high when two rare words occur mostly together. That is useful for names and technical compounds, and it is also why a frequency-1 accident can outrank city council. NPMI divides by the self-information of the pair, which reduces the extra reward for rarity and does not remove the reward for exclusivity.

Chi-square

$\chi^2 = N (O_{11}O_{22} - O_{12}O_{21})^2 / ((O_{11}+O_{12})(O_{11}+O_{21})(O_{12}+O_{22})(O_{21}+O_{22}))$.

$\chi^2_{Yates} = N (\max(0, |O_{11}O_{22} - O_{12}O_{21}| - N/2))^2$ / (the same denominator).

Chi-square is non-directional. It detects dependence, not attraction, and it does not penalize a high-frequency grammatical pair in the way PMI's ratio does. A perfectly exclusive pair of any positive count saturates at N, so the top of an unfiltered chi-square list is a tie, broken only by sorting the phrase string. Yates's continuity correction subtracts N/2 inside the deviation and floors at zero, which demotes

those exclusive hapaxes.

Student's t

The collocation form in Manning and Schütze (1999, eq. 5.10) is $t = (f(w_1 w_2) - f(w_1)f(w_2)/N) / \sqrt{f(w_1 w_2)}$. The denominator is the Poisson assumption that variance equals the mean, with the mean estimated by the observed bigram count.

That expression is the ordinary one-sample statistic $t = (\text{x-bar} - \mu) / \sqrt{(s^2 / n)}$ with $n = N$, $\text{x-bar} = O_{11}/N$, $\mu = P(w_1)P(w_2)$, and $s^2 = \text{x-bar}$. Manning and Schütze note that the normality assumption is not strictly satisfied by count data. The score is still used as a rank. It is directional, and it grows with frequency, so common grammatical pairs outrank rare domain terms.

Likelihood ratio

Dunning (1993): $G^2 = 2 \Sigma_{ij} O_{ij} \ln(O_{ij} / E_{ij})$, with $E_{ij} = (\text{row}_i \cdot \text{column}_j) / N$. The term with $O_{ij} = 0$ contributes 0. Expanded on the 2×2 table, $G^2 = 2 (L_{11} + L_{12} + L_{21} + L_{22})$ where $L_{ij} = O_{ij} \ln(O_{ij}/E_{ij})$ when $O_{ij} > 0$.

G^2 is asymptotically chi-square with one degree of freedom for a 2×2 table with fixed margins, the same reference distribution as Pearson's statistic. It is better behaved at the low counts typical of voice memos. Like chi-square, a large value means dependence, not specifically attraction.

Asymptotic equivalence does not imply the same top-20. It describes the null distribution as N grows, not a monotone relationship between the two scores on strongly associated pairs. On the 3864 candidates with frequency at least 3, the Spearman correlation of our chi-square and our G^2 is 0.9644, and the two top-20 lists share only *long beach*. An exclusive pair scores Pearson chi-square equal to the bigram N (61728), at any frequency. *boot camp* (frequency 4, both marginals 4) has chi-square 61728 and G^2 85.1533 (ranks 1 and 236). Decomposing Pearson's sum, the O^{11} term is 61720.0003 and the O^{22} term is 0.0003, so the huge complementary cell is not what puts exclusive names on top. $E^{11} = 0.000259$, and $(O^{11} - E^{11})^2 / E^{11}$ saturates at N. $G^2 = 2 \Sigma O^{ij} \ln(O^{ij} / E^{ij})$ is led by the O^{11} term, which grows with the joint count. *thank you* (frequency 488, marginals 524 and 1129, $O^{11}/E^{11} = 50.92$) has G^2 3890.29, of which 3835.90 is that O^{11} term, and chi-square 24535 (rank 25), under the exclusive-pair ceiling. The same pattern holds for the rest of each head: *coffee shop*, *dry ice*, *mm hmm*, and *san francisco* sit near N with G^2 between 66 and 153, while *want to*, *you know*, *i think*, and *do not* lead G^2 with chi-square ranks 128 to 154. Both formulas match scipy.stats.chi2_contingency on these tables (Pearson and Yates within 1e-11; the uncorrected log-likelihood statistic exactly). The N in the table is the bigram count, not the 67046 word tokens.

Trigram PMI

$\text{PMI}(w_1, w_2, w_3) = \log_2(f(w_1 w_2 w_3) \cdot N^2 / (f_1 f_2 f_3))$, where N is the number of sentence-internal trigram tokens and each f_i is the count of trigrams with that word in that slot.

Worked count table

The toy sentence is *city council city council budget*. It has 5 tokens and N = 4 bigrams. The pair under test is *city council*. $O^{11} = 2$, $O^{12} = 0$, $O^{21} = 0$, $O^{22} = 2$. $P(\text{pair}) = 0.5000$, $P(w^1) = 0.5000$, $P(w^2) = 0.5000$.

PMI = $\log^2((2/4) / ((2/4) \cdot (2/4))) = 1.000000$. NPMI = 1.000000. $\text{PMI}^2 = 1.000000$. $\text{PMI}^3 = 1.000000$.

$\chi^2 = 4 \cdot (4.0000)^2 / 16.0000 = 4.000000$. Yates uses $\max(0, |\text{cross}| - N/2) = \max(0, 4.0000 - 2.0000) = 2.0000$, so $\chi^2_{\text{Yates}} = 1.000000$.

The general t uses x-bar = 0.500000, $\mu = 0.250000$, $s^2 = \text{x-bar}$, $n = N$. It equals 0.707107, the same value as the Manning and Schütze form 0.707107. Expected cells under independence are all

1.0000. G^2 = 5.545177.

Calling NLTK's association functions on these same four cells reproduces PMI 1.000000, chi-square 4.000000, t 0.707107, and G^2 5.545177. Building an NLTK finder on the same five tokens sets finder.N = 5 and scores city council at PMI 1.321928, not at 1.000000. The gap is the token-count convention, not the logarithm.

pair	f	PMI	chi-square	Yates	t	G^2
city council	2	1.0000	4.0000	1.0000	0.7071	5.5452
council budget	1	1.0000	1.3333	0.0000	0.5000	1.7261
council city	1	1.0000	1.3333	0.0000	0.5000	1.7261

Every cell in that table was computed by src/models.py on the five-token sentence.

Preprocessing

Transcripts are read from data/raw_transcripts as plain text, one agenda item per file, in filename order. Unicode punctuation is normalized, sentences come from NLTK's sentence tokenizer, and tokens come from NLTK's word tokenizer. Tokens are lowercased. Numeric tokens and bare punctuation are dropped. Contractions that the tokenizer splits are repaired: n't becomes not, 're becomes are, and a bare 's is dropped because it is ambiguous between the possessive and the copula. Lemmas use WordNet with a part-of-speech tag from the averaged perceptron tagger. Bigrams and trigrams never cross a sentence boundary.

The main experiment keeps stopwords. The processed corpus has 67046 tokens, 5318 sentences, and 4437 lemmas. Removing stopwords would delete motion to, point of order, and in term of, and it would change N and both marginals, so every score would describe a different sequence from the one that was spoken. An ablation that does remove them leaves 31812 tokens. Only 2 gold bigrams contain a stopword (motion to, move to), but Student's t precision@20 moves from 0 on the main path to 0.550 after the deletion, because of the can no longer occupy the top of the list. That gain is the trade-off: fewer grammatical false friends, and every function-word formula destroyed.

Lemmatization has its own cost. is, are, and was become be, so this is becomes the pseudo-bigram this be. paid sick becomes pay sick. in terms of becomes in term of. The gold list uses the lemmatized string, because that is the string the ranker emits. Rankings that look topical and are actually copulas are called out in the evaluation, not quietly relabeled.

Pipeline

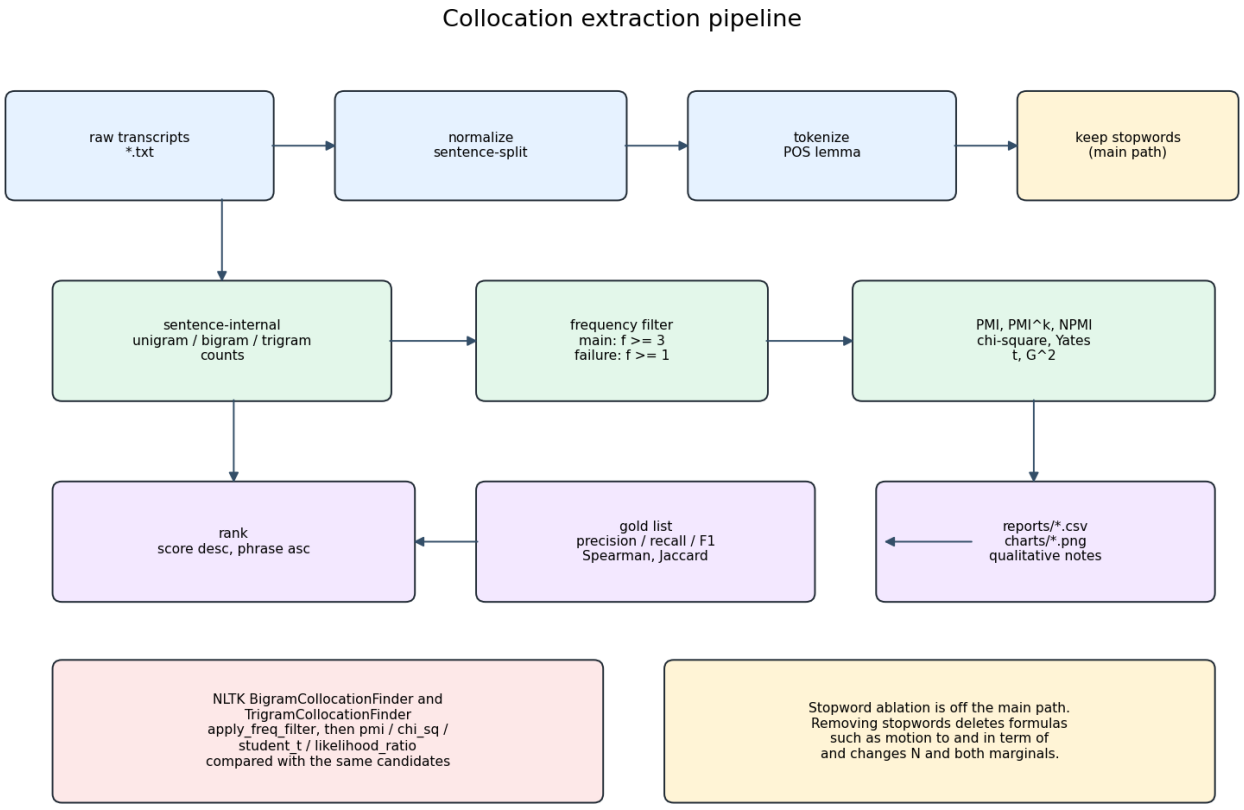


Figure. Main path, NLTK comparison, and the stopword ablation that is kept off that path.

The runner builds unigram, bigram, and trigram counts, scores every bigram at frequency at least 3, and repeats the ranking at frequency at least 1 for the low-count failure analysis. Candidates are sorted by score descending and phrase ascending, with no random sampling. Precision@k = |top k ∩ gold| / k, recall@k = |top k ∩ gold| / |gold|, and F1 is the harmonic mean of those two. Bigram metrics use the bigram gold list only (104 phrases). Trigram metrics use the trigram gold list only (40 phrases). Spearman correlation is computed on the intersection of candidates that every bigram metric scored. That intersection has 3864 phrases. Top-20 overlap is the Jaccard index and the raw intersection size.

NLTK beside the from-scratch scores

The assignment's library path is used as a second implementation, not as the definition of the math. BigramCollocationFinder.from_documents and TrigramCollocationFinder.from_documents are built on the same lemmatized sentences. apply_freq_filter(3) is applied for the main ranking. BigramAssocMeasures.pmi, chi_sq, student_t, and likelihood_ratio, and the trigram equivalents, produce the NLTK rankings.

On identical cells, the median ratio of NLTK PMI to our PMI is 1.000000 across 3864 filtered bigrams. A natural logarithm would have produced a ratio near 0.693147. The measured ratio matches log base 2 (log2). The median ratio of NLTK's likelihood ratio to our G^2 is 1.000000. A $\log^2$ implementation of G^2 would sit near 1.442695. The measured ratio matches the natural logarithm (ln), which is what Dunning's definition uses.

The median absolute difference between NLTK chi-square and our Pearson chi-square, on the same cells, is 0.000000. The median absolute difference against the Yates variant is 5.411539. NLTK therefore implements pearson no yates. The median absolute difference for the t statistic on the same cells is 0.000000. NLTK adds 1e-20 inside a square root and a logarithm; at these counts that smoother is invisible.

Finder scores still disagree with ours, because finder.N is the token count and the marginals are raw unigram counts rather than left and right bigram counts. The largest absolute PMI gap on the filtered set is yes thank: our score 5.8103 versus the finder score 2.7155. The largest chi-square gap is brook tucker (35271.4 versus 19500.9). Rank correlation between the two PMI lists remains high, which is a statement about order, not about the value of N.

Design limits

- The corpus is a 28-file subset of MeetingBank's test split, not the full set of council meetings, and not AMI or VoxPopuli audio.
- Contiguous sentence-internal windows miss collocations that are interrupted by a disfluency or a parenthetical.
- Lowercasing merges US and us. Lemmatization merges inflection and manufactures be-collocations.
- Chi-square and G^2 are used as ranks, not as tests with a reported p-value. The asymptotic chi-square reference is noted and not applied as a cutoff.
- The gold list is incomplete on purpose as a human sample. Precision@20 against it is a lower bound.
- No neural ranker and no part-of-speech filter on the candidate list. The comparison is among association scores.